Chapter 13 | Computer Graphics & Multimedia

13.1 Display Devices — CRT, Scanning and the Frame Buffer

(1) **The CRT is responsible for the actual display** in a computer — VRAM stores the image and the GPU computes it, but the cathode-ray tube generates the visible output. Inside the CRT, the **CONTROL GRID regulates the intensity of the electron beam** (brightness); the focusing and accelerating anodes shape and speed the beam. **Resolution = the maximum number of points that can be displayed WITHOUT OVERLAP on a CRT**; persistence is how long the phosphor glows.

(2) **Scanning**: **RANDOM (vector) scan operates like a pencil, drawing the line image directly** on the screen — the asked wording; raster scan instead sweeps pixel by pixel, row by row. Even when the screen is dark between frames, **commercial motion pictures use 24 frames per second, i.e. 72 screen illuminations per second** (each frame flashed three times to avoid flicker).

(3) **Emissive vs non-emissive displays**: plasma panels, thin-film electroluminescent displays and LEDs all EMIT their own light — the **Liquid Crystal Device is NOT emissive** (it modulates a backlight).

(4) **The frame buffer is the memory buffer that stores the image definition** — the intensity/colour of every pixel, read out to refresh the screen. **With ONE bit per pixel it is a BITMAP; with MULTIPLE bits per pixel it is called a PIXMAP** (the asked distinction). Frame-buffer size = resolution × bits per pixel.

Exam Points — 13.1

✦ CRT = the actual display; control grid = beam intensity; resolution = max non-overlapping points; random scan = pencil-like line drawing; film = 24 fps / 72 illuminations.

✦ LCD is the non-emissive one (plasma, electroluminescent, LED emit); frame buffer stores the image definition; 1 bit/pixel = bitmap, multiple bits = pixmap.

13.2 Aspect Ratio, Resolution and Size — Worked Problems

Aspect ratio: the ratio of WIDTH to HEIGHT (width always first). The **widescreen standard for monitors and TVs today is 16 : 9**.

All five asked numericals follow from one relation:

$$Aspect\ ratio = width \div height,\ so\ \ width = height \times ratio,\ \ height = width \div ratio$$

(1) Width 4 in, ratio 3 : 2 → **height = 4 × 2/3 = 8/3 inch**.

(2) Height 4 in, ratio 3 : 2 → **width = 4 × 3/2 = 6 inches**.

(3) Height 4.5 in, ratio 1.5 → **width = 4.5 × 1.5 = 6.75 inches**.

(4) **Resolution in pixels per inch = pixel count ÷ inches**: a 512 × 512 image of 2 × 2 inches → 512 ÷ 2 = **256 pixels per inch**.

File size (KVS): a 640 × 480 picture in 256 colours — 256 colours means 8 bits = 1 byte per pixel:

$$640 \times 480 \times 1\ byte = 307{,}200\ bytes \approx 300\ KB$$

Uncompressed bit rate (KVS): 800 × 600 pixels, 8 bits/pixel, 40 frames/second:

$$800 \times 600 \times 8 \times 40 = 153{,}600{,}000\ bps = 153.6\ Mbps$$

Note: That enormous uncompressed rate is exactly why MPEG compression exists.

Exam Points — 13.2

✦ Aspect ratio = width : height; widescreen = 16 : 9. Worked: 4 in wide at 3:2 → 8/3 in high; 4 in high → 6 in wide; 4.5 in at 1.5 → 6.75 in wide; 512 px over 2 in = 256 ppi.

✦ File size = pixels × bytes/pixel: 640×480 at 256 colours ≈ 300 KB. Bit rate = pixels × bits × fps: 800×600×8×40 = 153.6 Mbps.

13.3 Colour Models and Digital Image Processing

RGB and CMY are complements — each CMY component is one minus the matching RGB component, and vice versa:

$$CMY = 1 - RGB \qquad RGB = 1 - CMY$$

(1) RGB (0.2, 1, 0.5) → CMY **(0.8, 0, 0.5)** — the 2021 working.

(2) CMY (0.25, 0.65, 0) → RGB **(0.75, 0.35, 1)** — the second working.

(3) **Digitising an image**: sampling picks the positions; **QUANTIZATION digitises the AMPLITUDE value** (the asked term). **Digital Image Processing includes computer graphics, pixels and the camera mechanism — all of the mentioned**. In the standard pipeline, **the step after COMPRESSION is MORPHOLOGICAL PROCESSING**.

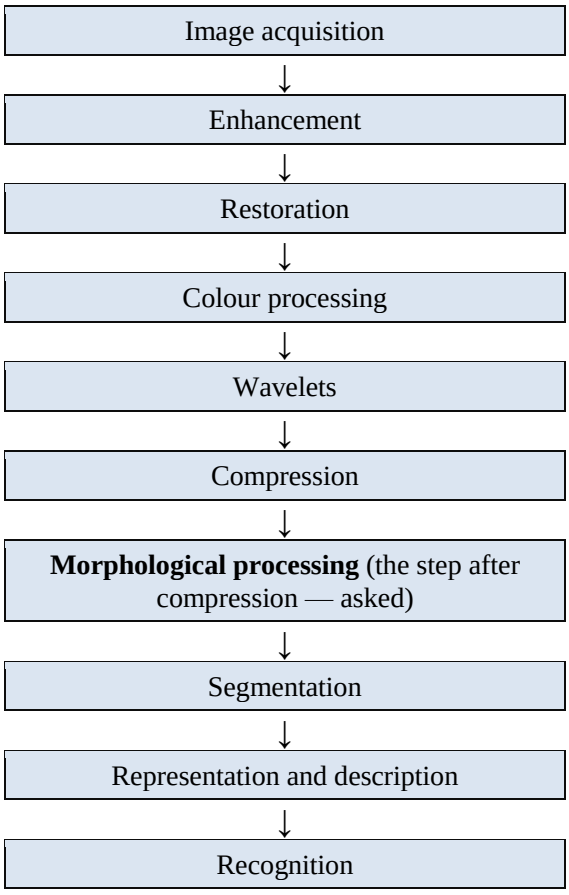


Fig. 13.1 — The digital image processing pipeline.

(4) **Pixel storage**: an image is a grid of rows and columns, so a **2-D array** is the natural store for pixel values. **Colour depth**: 'high color' images = **16 bits per pixel** (65,536 colours; 8-bit = 256 colours, 24-bit = true colour). **BMP supports 1/4/8/16/24/32-bit depths — 128-bit is NOT supported**. The CSS rule to make images fully grey: **img { filter: grayscale(100%); }** — the function is grayscale, on the filter property, unquoted.

Exam Points — 13.3

✦ CMY = 1 − RGB: (0.2,1,0.5) → (0.8,0,0.5); (0.25,0.65,0) → (0.75,0.35,1). Quantization digitises amplitude (sampling digitises position); DIP = all of the mentioned; after compression → morphological processing.

✦ Pixels live in a 2-D array; high color = 16 bpp; BMP has no 128-bit depth; img { filter: grayscale(100%); } for grey images.

13.4 Image Formats and Editing Tools

Format	Type	Asked facts
JPEG	Raster, lossy	**Joint Photographic Experts Group** — the full form asked TWICE (DSSSB 2021, Bihar 2023); best for photographs
PNG	Raster, lossless	**The common web-export format** and **the format supporting TRANSPARENCY** (alpha channel) — both asked
TIFF	Raster, lossless	**Tagged Image File Format** (full form asked); tags make it flexible for print/archives
BMP / DIB	Raster	**.dib = the Bitmap Image** (device-independent bitmap) — same raster data as .bmp
SVG	**Vector**	**The vector graphic format** in the PNG/GIF/JPG/SVG list — mathematical shapes, scales without loss
WAV	—	**NOT a graphics format** (audio) — the odd one out against GIF/JPG/PNG

(1) **Photoshop / editing tools** (2025 set): **brightness and contrast are adjusted from the ADJUSTMENTS PANEL**; the **cutting tool removes parts of the image**; the **CLONE STAMP duplicates part of the image to another area** (blemish and object removal). In Flash, the easiest way to draw a heptagon is the **PolyStar Tool with the 'Polygon' style** set to seven sides.

Exam Points — 13.4

✦ JPEG = Joint Photographic Experts Group (×2); PNG = web export + transparency; TIFF = Tagged Image File Format; .dib = bitmap image; SVG = vector; WAV not graphics.

✦ Adjustments panel = brightness/contrast; cutting tool removes parts; clone stamp duplicates areas; PolyStar (Polygon style) draws a heptagon in Flash.

13.5 Multimedia — Audio, Video, Animation and MPEG

(1) **Systems and tools**: multimedia **authoring software creates INTERACTIVE multimedia presentations** — **Adobe Flash is the authoring tool** in the Word/Photoshop/Flash list (editors handle one medium; authoring tools combine them). A **multimedia communication system facilitates the transmission of audio, video and data**. The **webcam is the device that captures video**; a **graphics package is a set of libraries giving programmatic access to 2D graphics functions** (OpenGL, DirectX). **Data compression reduces the size of multimedia files**; **digitization uses BOTH sampling and quantization**.

(2) **Animation**: it is **basically a form of PICTORIAL PRESENTATION** and **refers to SIMULATED MOTION pictures showing movement of drawn objects** (persistence of vision). To move objects and generate animation effects, **various TRANSFORMATION operations are necessary** (translation, rotation, scaling). **Morphing is the special effect changing one image into another through a seamless transition** (warp + cross-dissolve). **Frame rate is measured in FRAMES PER SECOND**; multimedia adds special **EFFECTS** to movies. **Kerning adjusts the space between CHARACTERS** in typography (leading = between lines). **Lip synchronization = synchronizing the audio and video streams**.

(3) **MPEG**: the standard is primarily a **BITSTREAM specification** — it fixes the decoder's input, leaving encoders free. The **macroblock layer = 16 × 16 pixels** (four 8 × 8 DCT blocks). MPEG uses spatial compression (JPEG-like, within a frame) and temporal compression, which **removes REDUNDANT frames** — successive frames repeat, so only differences are sent. The **MPEG audio extension is .mp3** (.mp4/.mov are video containers).

(4) Devices and views: the **handle-on-a-base controller is the JOYSTICK; the both-hands controller is the GAME PAD** — order (a) Joy-Stick, Game Pad in the HTET scenario. The **VANISHING POINT is where projected parallel lines appear to converge** (a perspective-projection property; parallel projection has none).

Exam Points — 13.5

- ✦ Authoring software = interactive presentations (Flash is one); communication system transmits A/V/data; webcam captures video; graphics package = 2D function libraries; compression shrinks files; digitization = sampling + quantization (both).
- ✦ Animation = pictorial presentation = simulated motion of drawn objects; needs transformation operations; morphing = seamless image-to-image; frame rate in fps; kerning = character spacing; lip sync = audio-video alignment.
- ✦ MPEG = bitstream spec; macroblock 16×16; temporal compression removes redundant frames; MPEG audio = .mp3. Joystick = handle on base, game pad = both hands; vanishing point = parallel lines converge.

13.6 Curves, Surfaces, Projection and Rendering

(1) Bezier curves: a Bezier curve is **a kind of spline generated from a set of CONTROL POINTS by polynomial (Bernstein) functions**, and it is a **PARAMETRIC curve defined by a minimum of three points — an origin, an endpoint and at least one control point** (both askings).

(2) Solid modeling is based on a **COMPLETE, VALID and UNAMBIGUOUS** geometric representation of the physical object. **When a plane is perpendicular to a reference plane, its projection on that plane is a STRAIGHT LINE** (seen edge-on). **Orthogonal projection and horizontal view cannot be framed together** for rendering — conflicting viewing geometries. **Rendering = producing bitmapped images from a view of 3-D models in a 3-D scene** (asked in the colour band).

(3) Lighting is based on the light source, the surface and the position of the observer — **the RENDERING ALGORITHM is NOT a lighting parameter** (it computes the result).

(4) Z-buffer algorithm (2023 statement set): it IS an image-space visible-surface method (i true); it uses a depth buffer for depths plus a refresh/frame buffer for intensities (ii true); it finds only one visible surface per pixel (iv true) — **the INCORRECT statement is iii only: object edges canNOT be anti-aliased** by the basic Z-buffer.

(5) Interpolation methods in graphics: Inverse Distance Weighted, Kriging, Natural Neighbor, Spline and Topo to Raster — **all five (i–v); 'Compiler' is not one**.

Stray (Chapter 11 topic): One ASP.NET item appears in this chapter's papers: **the System.Web.UI namespace defines the classes and interfaces for constructing and rendering elements on a web form** — covered with server-side technologies in Chapter 11.

Exam Points — 13.6

- ✦ Bezier = spline from control points; parametric, minimum three points (origin + endpoint + control). Solid modeling = complete/valid/unambiguous; perpendicular plane projects as a straight line; orthogonal + horizontal views cannot combine; rendering = bitmaps from 3-D scenes.
- ✦ Lighting parameters = source + surface + observer (not the rendering algorithm); Z-buffer: only 'edges can be anti-aliased' is false; interpolation = IDW/Kriging/Natural Neighbor/Spline/Topo-to-Raster (not Compiler).

13.7 2D / 3D Transformations

(1) Scaling multiplies each coordinate by its factor. The 2021 working — scale 2 in x and 4 in y on triangle A(0,0), B(3,3), C(2,5):

$$A(0,0) \rightarrow (0,0) \quad B(3,3) \rightarrow (6,12) \quad C(2,5) \rightarrow (4,20)$$

(2) Rotation about the origin by angle θ uses the standard matrix — with θ = 30° this was the asked matrix (key b):

$$R(\theta) = [\, cos\theta \quad -sin\theta \,;\; sin\theta \quad cos\theta \,] \Rightarrow R(30°) = [\, \sqrt{3}/2 \quad -1/2 \,;\; 1/2 \quad \sqrt{3}/2 \,]$$

(3) Surface of revolution (2021): the parametric curve x = cos(πt/2), y = sin(πt/2), 0 ≤ t ≤ 1 is a quarter-circle of radius 1; rotated 360° about the X-axis it sweeps a hemisphere, whose curved surface area is

$$2\pi r^2 = 2\pi(1)^2 = 2\pi$$

Exam Points — 13.7

✦ Scaling (2,4): B(3,3)→(6,12), C(2,5)→(4,20), origin fixed. Rotation matrix rows: cos −sin / sin cos. Quarter-circle revolved about X-axis → hemisphere, area 2π.

13.8 Clipping, Windowing and Raster Algorithms

(1) Clipping determines the visible and invisible portion of each element — visible kept, invisible discarded; it **can be applied through hardware as well as software**; and **point, area and text clipping are all types** — the 2023 triple is **all three TRUE**.

(2) Cohen–Sutherland line clipping: a line wholly inside is accepted, wholly outside rejected, and one that **INTERSECTS the clipping window is partially visible and must be clipped**. Each endpoint gets a **4-bit region code (Top, Bottom, Right, Left)**:

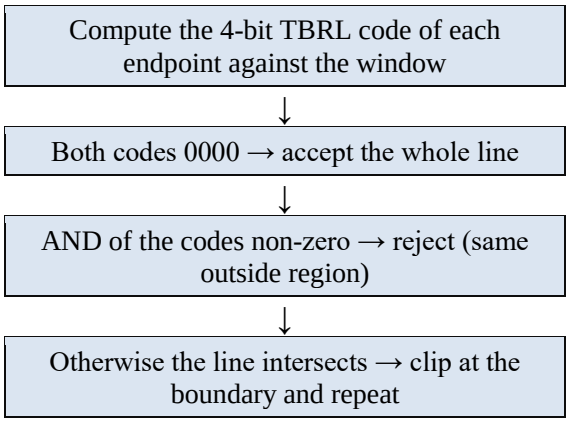


Fig. 13.2 — Cohen–Sutherland clipping. Worked (2021): window (−3,1)–(2,6); P(−4,2) lies LEFT → 0001; Q(−1,8) lies ABOVE → 1000.

(3) Which algorithm does what: **Liang–Barsky is the CLIPPING algorithm** in the list — Simple DDA, Bresenham and the mid-point algorithm are line-DRAWING (rasterisation) algorithms. A **circle has EIGHT-way symmetry** — one computed octant point yields seven reflections, which circle-drawing algorithms exploit.

(4) Aliasing (jagged edges) is reduced **by REDUCING the high-frequency components of the image by BLURRING it** — anti-aliasing is low-pass filtering before sampling.

Exam Points — 13.8

✦ Clipping triple all true (visible kept; hardware + software; point/area/text types). Cohen–Sutherland: intersecting line = partially visible; TBRL codes — P(−4,2) → 0001, Q(−1,8) → 1000.

✦ Liang–Barsky = clipping; DDA/Bresenham/mid-point = line drawing; circle = 8-way symmetry; aliasing cut by blurring away high frequencies.

Quick Revision — Chapter 13 (25 Points)

1. CRT = the actual display device; control grid regulates beam intensity; resolution = maximum points without overlap.

2. Random (vector) scan draws lines like a pencil; raster scan sweeps pixels row by row; film = 24 fps = 72 illuminations per second.

3. Emissive: plasma, electroluminescent, LED. Non-emissive: LCD (modulates a backlight).

4. Frame buffer stores the image definition; 1 bit/pixel = bitmap, multiple bits/pixel = pixmap; size = resolution × bits per pixel.
5. Aspect ratio = width : height (width first); widescreen standard = 16 : 9.
6. Aspect-ratio workings: 4 in wide at 3:2 → 8/3 in high; 4 in high at 3:2 → 6 in wide; 4.5 in at 1.5 → 6.75 in wide.
7. Resolution = pixels ÷ inches: 512 px over 2 in = 256 ppi. File size: 640×480 × 1 byte ≈ 300 KB. Bit rate: 800×600×8×40 = 153.6 Mbps.
8. CMY = 1 − RGB and RGB = 1 − CMY: (0.2,1,0.5)→(0.8,0,0.5); (0.25,0.65,0)→(0.75,0.35,1).
9. Sampling digitises position; QUANTIZATION digitises the amplitude value; pixels are stored in a 2-D array.
10. Image-processing pipeline: after compression comes morphological processing; DIP includes graphics, pixels and the camera (all).
11. High color = 16 bits/pixel; 8-bit = 256 colours; 24-bit = true colour; BMP never 128-bit.
12. JPEG = Joint Photographic Experts Group (lossy, photos); PNG = web format with transparency; TIFF = Tagged Image File Format; .dib = bitmap; WAV is audio, not graphics.
13. SVG is the vector format — scales without loss; PNG/GIF/JPG are raster.
14. Photoshop: Adjustments panel (brightness/contrast), cutting tool (removes parts), clone stamp (duplicates areas); Flash PolyStar draws polygons.
15. Authoring software builds interactive presentations (Adobe Flash); a multimedia communication system transmits audio, video and data; webcam captures video.
16. Compression shrinks multimedia; digitization = sampling + quantization; graphics package = 2D function libraries.
17. Animation = pictorial presentation of simulated motion; requires transformation operations; morphing blends one image into another; frame rate in frames per second.
18. Kerning = space between characters; lip synchronization aligns audio with video; multimedia adds special effects to movies.
19. MPEG = a bitstream specification; macroblock = 16×16 pixels; temporal compression removes redundant frames; MPEG audio = .mp3.
20. Joystick = handle on a base; game pad = held in both hands. Vanishing point = where parallel lines converge (perspective only).
21. Bezier curve = parametric spline from control points (origin + endpoint + at least one control point, minimum three).
22. Solid modeling = complete, valid, unambiguous; a perpendicular plane projects as a straight line; lighting depends on source + surface + observer (not the algorithm).
23. Z-buffer: image-space method, depth + frame buffers, one surface per pixel — but it cannot anti-alias edges (the false claim).
24. Scaling multiplies coordinates: (3,3)→(6,12), (2,5)→(4,20) at scale (2,4); rotation matrix = cos −sin / sin cos; revolved quarter-circle = hemisphere, area 2π.
25. Clipping keeps the visible portion (hardware or software; point/area/text types); Cohen–Sutherland TBRL codes (left = 0001, top = 1000); Liang–Barsky clips while DDA/Bresenham/mid-point draw; circles have 8-way symmetry; anti-aliasing = blur away high frequencies.

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